

DAY-1

Introduction to WAN Technologies Theory :WAN vs. LAN: Key differences and use cases. MPLS basics. Practical :Observe MPLS routing in a pre-configured network.

Task: Document MPLS benefits over traditional routing.

[Q-1] Create a comparison table listing at least 3 key differences between WAN and LAN, and give one real-world example of each from apps or services you use daily (for example, IRCTC, Instagram, or Zomato).

(A)

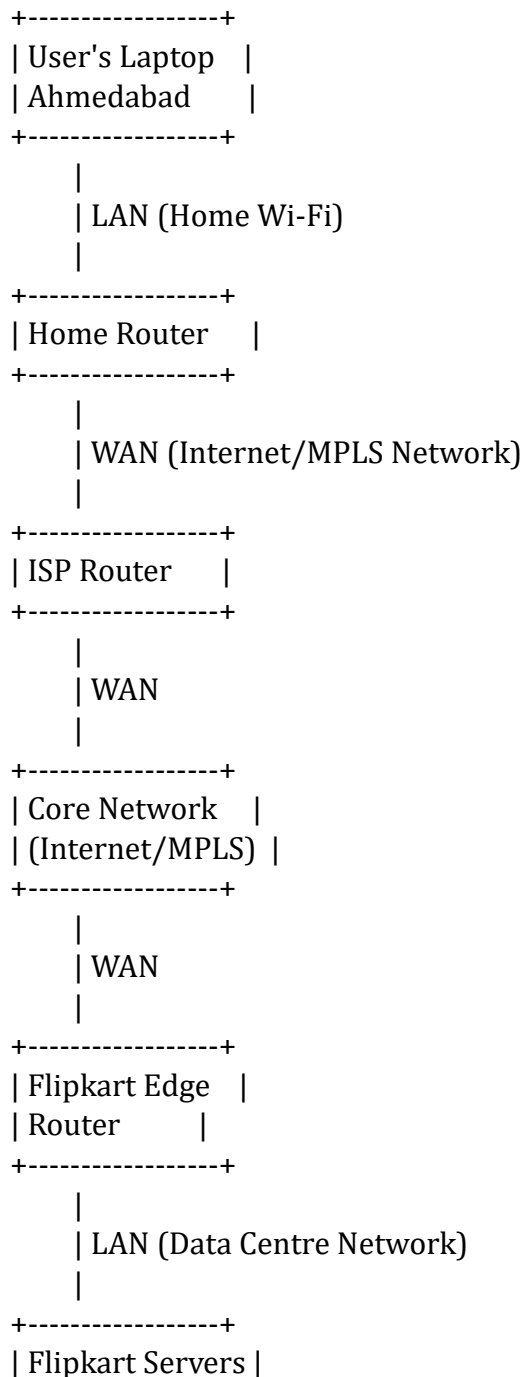
Feature	LAN (Local Area Network)	WAN (Wide Area Network)
Coverage Area	Covers a small area such as a home, office, or school.	Covers a large geographical area such as cities, states, or countries.
Speed	Usually faster because devices are connected locally.	Usually slower than LAN due to long-distance communication.
Ownership	Owned and managed by a single organization or individual.	Often uses service providers (ISPs) and multiple networks.
Cost	Less expensive to set up and maintain.	More expensive due to infrastructure and service provider costs.

Real-World Examples

- **LAN Example:** A home Wi-Fi network connecting a laptop and smartphone to access **Instagram**.
- **WAN Example:** Accessing **IRCTC**, **Flipkart**, or **Zomato**, where users connect through the Internet to servers located in different cities.

[Q-2] Draw a simple network diagram using any drawing tool (or on paper and upload a photo) that shows how a user in Ahmedabad could connect to a Flipkart data centre in Bangalore using a WAN, labelling the LAN and WAN segments clearly.

(A)



| Bangalore |
+-----+

Labels:

- LAN Segment: User Laptop ↔ Home Router
- WAN Segment: Home Router ↔ ISP ↔ Core Network ↔ Flipkart Router

[Q-3] Research and write a short summary (5-7 lines) explaining what MPLS is and how it improves data routing compared to traditional IP routing.

(A)

MPLS (Multiprotocol Label Switching) is a high-speed networking technology used to forward packets using labels instead of examining IP addresses at every router. When a packet enters an MPLS network, a label is attached to it. Routers then forward the packet based on this label, making routing decisions faster. MPLS reduces the processing load on routers and improves network efficiency. It also supports traffic engineering, allowing network administrators to choose the best path for data. As a result, MPLS provides better performance, reliability, and Quality of Service (QoS) compared to traditional IP routing.

[Q-4] Observe the pre-configured MPLS network in the provided Packet Tracer or GNS3 file, and note down two routing behaviours that are different from standard IP routing.
Hint: Look for how labels are used and how packets move between routers.

(A)

1. Label-Based Forwarding

- In traditional IP routing, every router checks the destination IP address and routing table.
- In MPLS, routers forward packets using labels attached to the packets.
- This speeds up packet forwarding.

2. Predefined Label Switched Paths (LSPs)

- Traditional IP routing selects the next hop at each router independently.

- MPLS uses predefined paths called Label Switched Paths (LSPs).
- Packets follow the established path, resulting in more predictable and efficient routing.

[Q-5] List three specific benefits MPLS offers over traditional routing for companies running high traffic apps like YouTube or BookMyShow, and briefly explain how each benefit helps in real world scenarios.

(A)

1. Faster Data Forwarding

Benefit: MPLS uses labels instead of repeated IP lookups.

Real-World Example:

YouTube videos load and stream more smoothly because packets are forwarded more quickly through the network.

2. Better Traffic Management (Traffic Engineering)

Benefit: Network administrators can control the routes taken by different types of traffic.

Real-World Example:

BookMyShow can prioritize ticket booking traffic during a major event release, reducing delays for users.

3. Quality of Service (QoS)

Benefit: MPLS can prioritize important traffic such as video, voice, or critical business applications.

Real-World Example:

During live YouTube streaming, video traffic can be given higher priority, reducing buffering and improving viewing quality.